Purely Analytic Closure of the Final Spherical-Shell Residual

Abstract

This note assembles the analytic compact and aggregate estimates after the exact owner subtraction. It proves that the final spherical-shell residual is empty without invoking an interval certificate or an executable truth table. The compact estimate covers a closed superset of the actual sub-200 residual; the aggregate estimate covers the complementary frequency tail.

1 Notation and analytic inputs

Let

$$A_{\rho,1} := \{x \in \mathbb{R}^3 : \rho < |x| < 1\}, \qquad W(\rho, K) := \frac{2(1-\rho^3)K^3}{9\pi},$$

and let $N_D(A_{\rho,1}, K^2)$ count Dirichlet eigenvalues strictly below K^2. Put

$$\rho_c := \frac{1}{1+2\pi}, \qquad k_{11}(\rho) := \sqrt{\frac{\pi^2}{(1-\rho)^2} + 132}.$$

The preceding analytic owner lemmas and the exact set subtraction give

$$\mathcal{D}_{20} = \left\{(\rho, K) : \rho_c \le \rho < \frac{39}{50}, \quad k_{11}(\rho) < K < U(\rho)\right\}. \tag{1}$$

On this interval $U(\rho) = K_0(\rho)$, although the formula for the upper wall will not be needed below.

We use the shell action

$$G_a(z) := \frac{1}{\pi}\left(\sqrt{a^2 - z^2} - z \arccos \frac{z}{a}\right) \quad (0 \le z < a), \qquad G_a(z) := 0 \quad (z \ge a),$$

$$A_{\rho,K}(z) := G_K(z) - G_{\rho K}(z),$$

and the strict lattice proxy

$$P(\rho, K) := \sum_{\substack{\nu=\ell+1/2<K \\ \ell \in \mathbb{N}_0}} 2\nu \, \# \left\{n \in \mathbb{N} : n < A_{\rho,K}(\nu) + \frac{1}{4}\right\}. \tag{2}$$

The analytic Bessel-phase reduction proved earlier in the main argument is

$$N_D(A_{\rho,1}, K^2) \le P(\rho, K). \tag{3}$$

The detailed analytic supplement establishes the following two estimates. They are stated here in exactly the form used in the assembly:

$$P(\rho, K) < W(\rho, K) \qquad \left(\rho_c \le \rho \le \frac{39}{50},\ k_{11}(\rho) \le K \le 200\right), \tag{4}$$

$$N_D(A_{\rho,1}, K^2) < W(\rho, K) \qquad \left(\rho_c \le \rho \le \frac{39}{50},\ K \ge 200\right). \tag{5}$$

For clarity, the proof of (4) is recalled in the next section. The proof of (5) is the analytic curvature propagation from the three exact base inequalities at $K = 200$.

2 Compact closure

Theorem 1 (Analytic compact theorem)**.** *For*

$$\rho_c \le \rho \le \frac{39}{50}, \qquad k_{11}(\rho) \le K \le 200,$$

one has

$$N_D(A_{\rho,1}, K^2) < W(\rho, K).$$

Proof. The retained-remainder identity in the analytic compact argument gives

$$W - P = \mathcal{B} + \mathcal{R}_{\mathrm{dec}} + \mathcal{R}_{\mathrm{osc}} - \mathcal{E}_{\mathrm{ang}},$$

with both retained remainders nonnegative. A global primitive minorant then proves the structural implication

$$\mathcal{E}_{\mathrm{ang}}(\rho, K) < \mathcal{H}(\rho, K) \quad \Longrightarrow \quad P(\rho, K) < W(\rho, K). \tag{6}$$

The paired angular-block lemma, including all common radial–angular jumps, gives

$$\mathcal{E}_{\mathrm{ang}}(\rho, K) < \frac{(1-\rho^2)K^2}{6} - \frac{1}{2}. \tag{7}$$

The scalar positivity theorem proves on the same closed domain that

$$\mathcal{H}(\rho, K) - \frac{(1-\rho^2)K^2}{6} + \frac{1}{2} > 0. \tag{8}$$

Combining (7) and (8) yields $\mathcal{E}_{\mathrm{ang}} < \mathcal{H}$. Hence (6) gives $P < W$, and (3) completes the proof. □

3 Exact compact/tail ownership

Proposition 2 (Analytic closure of $\mathcal{D}_{20}$)**.** *Define*

$$\mathcal{C}_{21} := \mathcal{D}_{20} \cap \{K \le 200\}, \qquad \mathcal{T}_{21} := \mathcal{D}_{20} \cap \{K > 200\}.$$

Then $\mathcal{C}_{21}$ and $\mathcal{T}_{21}$ are disjoint, exhaust $\mathcal{D}_{20}$, and the strict Pólya inequality holds on both. Consequently the successor residual is empty.

Proof. The two frequency conditions $K \le 200$ and $K > 200$ form a disjoint partition, so

$$\mathcal{D}_{20} = \mathcal{C}_{21} \,\dot{\cup}\, \mathcal{T}_{21}.$$

If $(\rho, K) \in \mathcal{C}_{21}$, then (1) gives

$$\rho_c \le \rho < \frac{39}{50}, \qquad k_{11}(\rho) < K \le 200.$$

This lies in the closed domain of Theorem 1.

If $(\rho, K) \in \mathcal{T}_{21}$, then

$$\rho_c \le \rho < \frac{39}{50}, \qquad K > 200,$$

so the analytic aggregate estimate (5) applies. Therefore every point of $\mathcal{D}_{20}$ is proved, and

$$\mathcal{D}_{20} \setminus (\mathcal{C}_{21} \cup \mathcal{T}_{21}) = \varnothing.$$

The face $K = 200$ is assigned only to $\mathcal{C}_{21}$; the already owned faces $\rho = 39/50$, $K = k_{11}(\rho)$, and $K = U(\rho)$ remain outside $\mathcal{D}_{20}$. □

Logical dependency. The proof uses finite exact-rational tables only through displayed analytic lemmas. It does not use the former 10,580-box compact certificate, the 1,286-box aggregate certificate, or the 63-state executable subtraction table.